Network Visualization Tools to Enhance Metabolic Engineering Platforms

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Abstract. In this work, we present a software platform for the visualization of metabolic models, which is implemented as a plug-in for the open-source metabolic engineering (ME) platform OptFlux. The tools provided by this plug-in allow the visualization of the models (or parts of the models) combined with the results from operations applied over these models, mainly regarding phenotype simulation, strain optimization and pathway analysis. The tool provides a generic input/ output framework that can import/ export layouts from different formats used by other tools, namely XGMML and SBML. Thus, this work provides a bridge between network visualization and ME.

Keywords: Metabolic models, biological network visualization, metabolic engineering, open-source software.

1 Introduction

Bioinformatics has recently become an important discipline in life sciences research, mainly due to the high amount of experimental data generated by high throughput sequencing and omics approaches. These data are stored in different databases, in multiple formats, bringing problems in their integration. Due to the difficulty of interpreting and analysing large-scale databases, networks became useful tools for understanding the complexity of biological processes, being commonly used to represent biological entities and their interactions. Indeed, networks are used to represent metabolic pathways, regulatory information and signal transduction systems[3].

Over the years, several tools arose to address the complexity of analysis and visualization tasks for large-scale networks. The use of graphs is very common, due to the easy representation of biological entities as nodes and their interactions as edges. The fact that each entity may be connected with several others highlights the importance of the multi-edged networks approach that several tools offer [10]. The main goal of any visualization tool is the capability of identifying or highlighting patterns and other information not easily available through the analysis of the raw data.

The ability to visually represent metabolic networks is already provided by several tools whose enumeration is out of the scope of this paper. In this regard, we would

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emphasize *Cytoscape* [14] that has emerged as a powerful and extendible tool for network analysis and visualization. While these tools already exist, they are not currently connected with any of the few existing Metabolic Engineering (ME) frameworks.

Some related efforts in the modelling of biological systems have been key tasks in the computational systems biology field, whose aim is to develop and use algorithms, data structures and tools to allow the integration of data into models that can be used to simulate the behaviour of biological systems. In particular, the development of metabolic models and tools to simulate the behaviour of these systems, allows for the creation of important tools in the context of ME, whose main goal is to optimize cellular metabolism deriving rational genetic modifications that lead to strains capable of producing compounds of interest.

While for the dynamic modelling of large-scale systems, it is often necessary to have mechanistic detail and kinetic parameters not commonly available, structured-oriented analyses only requires the usually well-characterized network topology [15]. Constraint-based analysis methods that consider network stoichiometry, and potentially other constraints such as maximal pathway capacities, are proving to be very reliable ways to obtain a quantitative perspective of various aspects of cellular function. In particular, these have gained considerable popularity for simulating cellular metabolism, using methods such as flux balance analysis (FBA) [11] to provide phenotype simulation and as the basis for strain optimization [13].

The use of these methods in the context of ME has provided a number of valuable algorithms and tools. The authors research group has been active in this line of research and the main results are available for the ME community in the form of an open-source software platform, *OptFlux* (`http://www.optflux.org`) [12]. This application includes a number of tools for ME, spanning phenotype simulation methods for wild type and mutant strains, strain optimization algorithms, support for several model formats, pathway and topological analysis, among others.

In this context, the main aim of this work is to develop computational tools for the visualization of metabolic networks, in the context of a metabolic engineering platform (*OptFlux*), providing an adequate integration between the features of both tools. The overall idea is to provide visualization capabilities to *OptFlux*, powering this tool with extended visual analysis capabilities. In more detail, the following technological objectives will be pursued:

- Implement a generic library for the visualization of metabolic models or parts of these models;
- Implement tools to allow importing/exporting model layouts in standard formats, such as *SBML* [8] and *XGMML* [2];
- Integrate those tools in a ME platform, integrating data obtained from phenotype simulations, strain optimization and pathway analysis, and allowing the exportation of the results in the standard formats.

2 Implementation

Metabolic models collect relevant metabolic information in a mathematical format. ME makes use of these models to predict ways to maximize the production of desired

compounds, through directed genetic changes. Usually, this task is done through phenotype simulations and maximizing desired fluxes from the reactions in the model. The visual representation of these networks should be as simple as possible, to facilitate naked-eye interpretation. Considering the nature of ME simulations, which generate flux values distributions, for given conditions, the network can then be viewed as a set of reactions, and a simulation results as a set of values for those reactions.

Taking this into account, a visual representation of a metabolic model was envisioned. As expected, the network is represented as a graph, where biological entities (reactions and metabolites) are represented as nodes, and their interactions (consumption and production) as edges. The architecture of the visualizer library has two distinct layers (Figure 1): the *BiovisualizerCore* contains the specification of a visual representation of the metabolic network (layout), the tools to represent it in a 2D interface and methods for visual manipulation of the network; the second layer, the Input/Output layer, consists in the readers and writers for a variety of formats that can provide layouts to the *BiovisualizerCore*, working as a transformer receiving a file in a specific format, reading it and generating a layout.

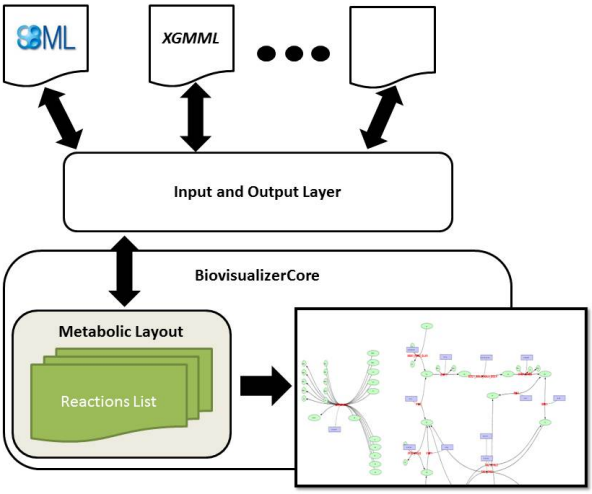


Fig. 1. Visualizer architecture: I/O layer imports/ exports standard network files and transforms them into metabolic layouts; the *BiovisualizerCore* layer has the capability to visually represent the metabolic layouts.

2.1 BioVisualizerCore

The *BiovisualizerCore*'s main purpose is to transform a metabolic layout into a 2D graph. The metabolic layout definition is represented by a list of reactions and was implemented as a Java interface. Each reaction has its identifier, spatial coordinates (x,y), reversibility information, additional information, a list of reactants and a list of

products. Each metabolite has information on its identifier and coordinates. Therefore, there are three types of nodes: reaction, metabolite and information nodes. The visual representation of the network is handled by the *Prefuse* library, a graph visualization framework written in Java [1].

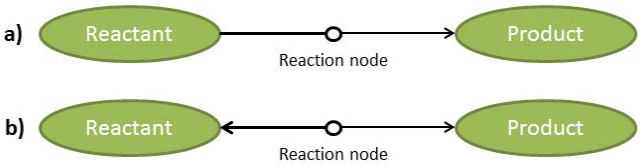


Fig. 2. *BiovisualizerCore*s representation of metabolic reactions. Each reaction has a reaction node, and reactants and products: a) irreversible reaction; b) reversible reaction.

Each node of the graph can have its 2D coordinates defined, but they are not mandatory. For the nodes that do not have a position, an automatic layout provided by *Prefuse* is applied, called *Force Directed Layout*, that positions graph elements based on a physics simulation of interacting forces; by default, nodes repel each other, edges act as springs, and drag forces are applied. *BiovisualizerCore* packages provide a series of functionalities to enhance the interpretation of the network. Hovering the cursor over a node highlights it; if the node is a reaction, it will also highlight its edges and neighbour nodes. It has a series of visual filters, allowing to change the visibility of nodes/ edges based on attributes. Finally, it also provides ways to change the thickness of edges based on numerical values.

2.2 Input/Output Layer

BiovisualizerCore gives the possibility to visually represent a metabolic layout, but building that layout is still needed. There are several tools that allow the creation of networks and exportation of those networks for several file formats. The I/O Layer has the objective of providing the capability to read a given network in a specific file format, and building the metabolic layout, readable by *BiovisualizerCore* packages. At the same time it shall also provide the possibility to export those metabolic layouts into those specific formats.

Currently, this layer supports two formats: *CellDesigner SBML* and *XGMML*. *CellDesigner* [6] is a well-known tool for drawing cellular networks, based on the process diagram paradigm, with a graphical notation system proposed by Kitano. These are stored using an extension of the *Systems Biology Markup Language (SBML)*. On the other hand, *XGMML (eXtensible Graph Markup and Modeling Lan-guage)* [2] is an extension to *GML (Graph Markup Language)*, and it is used for graph description

using XML tags to describe nodes and edges of a graph, being supported for instance by *Cytoscape*.

2.3 OptFlux Interaction

BioVisualizerCore packages are integrated within *Optflux* to provide new plug-ins for this framework. In terms of its implementation, *Optflux* is built on top of *AIBench* [7], a framework for the development of scientific applications, bringing important advantages to both the developers and the users, given its design principles and archi-tecture. The applications follow the MVC (model-view-controller) pattern and they are plug-in based facilitating the reuse and integration with additional developments. We designed an intuitive connection between layouts and metabolic models, map-ping identifiers of the nodes with identifiers of the respective compounds/reactions in the model. Note that the layouts can contain only a part of the entities in the model, being also possible to create distinct layouts for a given model.

Using this connection it will be possible to access metabolic information directly from the visualizer by clicking the nodes. Visual filters include hiding information nodes, hiding fluxes not present in the metabolic model and hiding reactions with zero value fluxes, given a flux distribution. Flux distributions can be obtained from phenotype simulations performed in *OptFlux*, bringing the capability to visualize the effect that those simulations have on the network, by changing the thickness of the edges of the reactions according to that simulation, by normalizing the value of the fluxes. The *I/O Layer* supports distinct file formats, and to integrate it with *OptFlux* and facilitate the addition of new formats, we decided to implement a series of *OptFlux* plugins. Currently, there are two plugins for the visualization of metabolic networks:

- *CellDesigner plugin*: supports *CellDesigner SBML* created layouts. The SBML reader used is the same *OptFlux* uses to read SBML models using JSBML, a Java library for reading, writing, and manipulating SBML files and data streams [4].
- *XGMML plugin*: gives support for networks created in the XGMML format. *Cytoscape* [14] is one of the tools that support the creation of these networks. The exportation works in two different ways: in the first, the layout imported is preserved, while in the other one exports the *BiovisualizerCore* layout appearance.

3 Case Study

To illustrate some of the features of the tool, we loaded a simplified model for Saccharomyces cerevisiae [5]. A *CellDesigner* SBML layout was also loaded.

Figure 3a) shows the overview of the network, with all nodes visible; 3b) shows the highlighted biomass reaction, with all its reactants and products also highlighted; 3c) shows metabolite CO2 highlighted, and since this metabolite appears in several reactions, those nodes are also highlighted; finally, 3d) shows a flux distribution obtained from simulating a wild type strain. The thickness of the edges is changed, the filter to hide zero value fluxes is activated and the labels of the reactions are changed by adding the value of the flux after the reaction name.

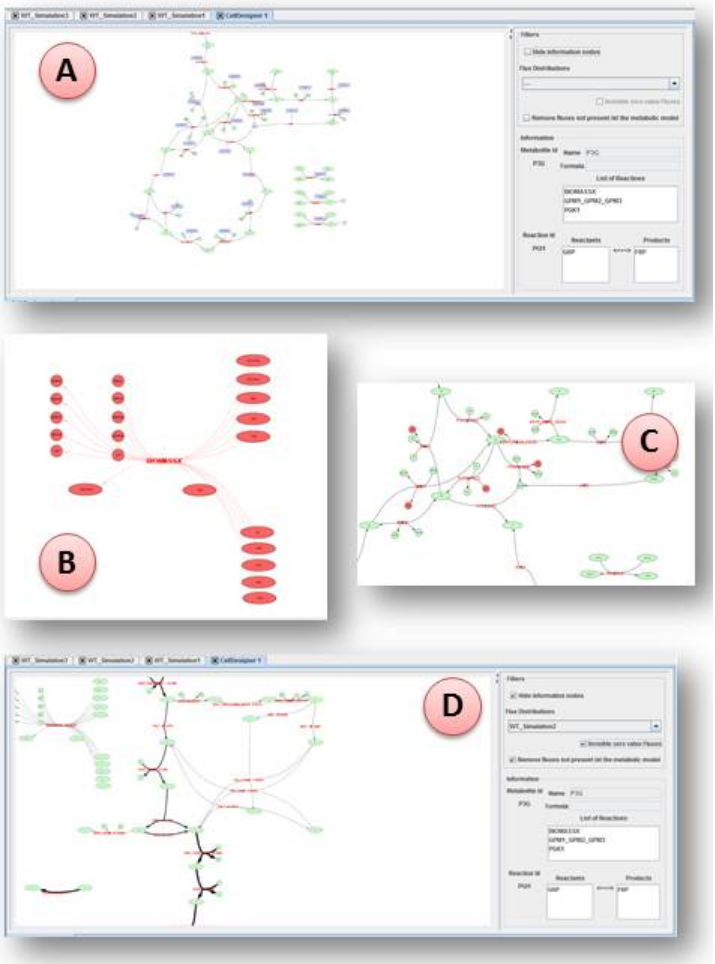


Fig. 3. Some functionalities of the visualizator. A. network overview; B. biomass reaction highlighted; C metabolite CO2 highlighted; D Flux distribution.

4 Conclusions and Future Work

BiovisualizerCore packages and *Optflux* plugins provide a freely available tool that allows users to visualize the results of *in silico* phenotype simulations, which is something new in ME tools. The ability to export those results and import them in other tools (e.g. *Cytoscape)* is also a valuable asset. One of the major applications of these features is the enhanced ability to analyse results of strain optimization results. In this scenario, these tools allow looking at simulations of the best mutants found by the strain optimization process and analysing their phenotype, thus visually searching for clues that can highlight the strategies followed to produce a given compound.

Also, these tools are linked with the elementary modes analysis plugin (EFM4OptFlux) providing a visualization tool for the set of elementary modes found for a given network, which can be then be exported for instance to *CellDesigner* or *Cytoscape*.

There are some aspects of the visualizer that can be improved in the future. Allowing the users a broader configuration of the aspect of the network is something worth exploring. Improving the I/O layer by providing a larger spectra of format support is another, for instance integrating KEGG [9] layouts through the format KEGG-ML.

OptFlux is freely downloadable from the website `http://www.optflux.org` and both plugins cited in this work are available through the plugin manager the software provides in the *Help* menu.

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